3 (a)	Simple harmonic motion is defined as the motion of an object whose acceleration *a* is **<u>proportional</u>** to its displacement *x* from a fixed point (equilibrium position)	B1
	and is always **<u>directed towards</u>** that fixed point.	B1
(b)(i)	forces in springs are $k(e + x)$ and $k(e - x)$ resultant = $k(e + x) - k(e - x)$ $= 2kx$	C1
	By Newton's Second Law, $F_{net} = ma$	M1
	negative sign explained as the acceleration is always oppositely directed to the displacement. $a = -\frac{2kx}{m}$	M1 A0
(b)(ii)1	$\omega^2 = \frac{2k}{m}$ $(2\pi f)^2 = \frac{2 \cdot (120)}{0.900}$	

		C1
	f = 2.60 Hz	A1
(b)(ii)2	$a_0 = \omega^2 x_0 = 5.2$ $x_0 = \dfrac{5.2}{\dfrac{2\cdot(120)}{0.900}}$	C1
	$= 1.95 \times 10^{-2}$ m or 1.95 cm	A1
(b)(ii)3	Max kinetic energy = $\frac{1}{2} m(\omega x_0)^2$ $= \frac{1}{2}(0.900)\left(\frac{2\cdot(120)}{0.900}\right)(0.0195)^2$	M1
	= 0.0456 J	A1

4 (a)	[illegible]	C1